MAT215: Complex Variables And Laplace Transformations

Emon Hossain[1]

[1]Lecturer
MNS department
Brac University

LECTURE-09

Definition of Harmonic Functions

Definition

A function $u(x, y)$ is called a *harmonic function* if it satisfies the Laplace's equation:

$$\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0.$$

Example

Example

Show that the function $u(x,y) = 3x^2y + 2x^2 - y^3 - 2y^2$ is a harmonic function.

Example

Show that the function $u(x,y) = \tan^{-1}\left(\frac{y}{x}\right)$ is a harmonic function.

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \stackrel{?}{=} 0$$

Given that, $u(x,y) = 3x^2y + 2x^2 - y^2 - 2y^2$

$$\frac{\partial u}{\partial x} = 3y \times 2x + 2 \times 2x - 0 - 0 = 6xy + 4x$$

$$\frac{\partial^2 u}{\partial x^2} = 6y + 4$$

$$\frac{\partial u}{\partial y} = 3x^2 + 0 - 3y^2 - 4y$$

$$\frac{\partial^2 u}{\partial y^2} = 0 - 6y - 4 = -6y - 4$$

$$\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 6y + 4 - 6y - 4 = 0$$

2nd $\quad u(x,y) = \tan^{-1}\left(\frac{y}{x}\right)$ $\qquad \frac{d}{dx}(\tan^{-1}x) = \frac{1}{1+x^2}$

$$\frac{\partial u}{\partial x} = \frac{\partial}{\partial x}\left(\tan^{-1}\left(\frac{y}{x}\right)\right) =$$

$$= \frac{1}{1+\left(\frac{y}{x}\right)^2} \times \frac{\partial}{\partial x}\left(\frac{y}{x}\right)$$

$$= \frac{1}{\frac{x^2+y^2}{x^2}} \times y \times -\frac{1}{x^2} = \frac{x^2}{x^2+y^2} \times y \times \frac{1}{-x^2}$$

$$= \boxed{\frac{-y}{x^2+y^2}}$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{(x^2+y^2)\frac{\partial}{\partial x}(-y) - (-y)\frac{\partial}{\partial x}(x^2+y^2)}{(x^2+y^2)^2} = \frac{0 + y \times 2x}{(x^2+y^2)^2}$$

$$= \frac{2xy}{(x^2+y^2)^2}$$

$$\frac{\partial u}{\partial y} = \frac{1}{1+\left(\frac{y}{x}\right)^2} \times \frac{\partial}{\partial y}\left(\frac{y}{x}\right) = \frac{1}{\frac{x^2+y^2}{x^2}} \times x \times \frac{1}{x}.$$

$$= \boxed{\frac{x}{x^2+y^2}}$$

$$\frac{\partial^2 u}{\partial y^2} = \frac{(x^2+y^2)\frac{\partial}{\partial y}(x) - x\frac{\partial}{\partial y}(x^2+y^2)}{(x^2+y^2)^2} = \boxed{\frac{-2xy}{(x^2+y^2)^2}}$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{2xy}{(x^2+y^2)^2} - \frac{2xy}{(x^2+y^2)^2} = 0.$$

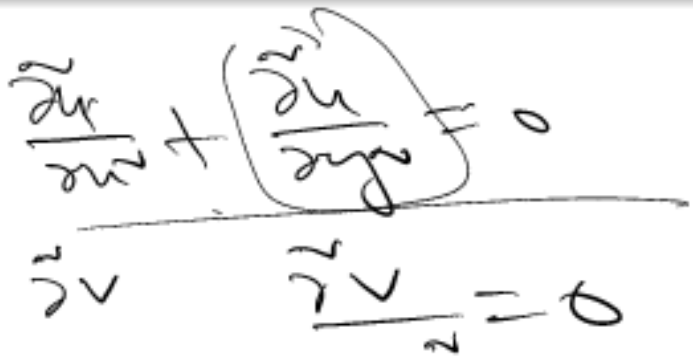
$u(x,y)$ is a harmonic function.

Theorem



Theorem

If $f(z) = u(x, y) + iv(x, y)$ is an analytic function in a region D, then both u and v are harmonic functions if they have continuous second partial derivatives in D.



Proof: Given that, $f(z)$ is analytic.

$$f(z) = u(x,y) + i\,v(x,y)$$

C-R Equations gives,

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad (1)$$

&

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \quad (2)$$

$$\frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\right) = \frac{\partial}{\partial x}\left(\frac{\partial v}{\partial y}\right)$$

$$\Rightarrow \frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 v}{\partial x \partial y}$$

&

$$\frac{\partial}{\partial y}\left(\frac{\partial u}{\partial y}\right) = \frac{\partial}{\partial y}\left(-\frac{\partial v}{\partial x}\right)$$

$$\frac{\partial^2 u}{\partial y^2} = -\frac{\partial^2 v}{\partial y \partial x}$$

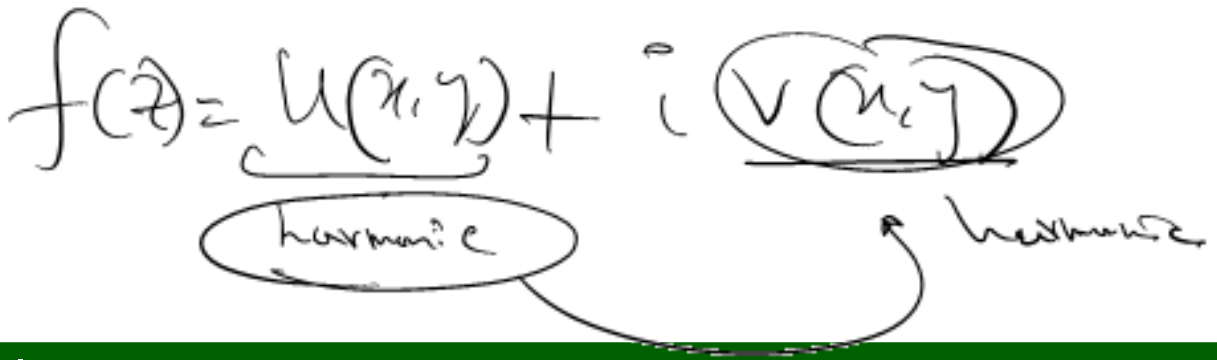
$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = \frac{\partial v}{\partial x \partial y} - \frac{\partial v}{\partial y \partial x} = \frac{\partial v}{\partial x \partial y} - \frac{\partial v}{\partial x \partial y} = 0$$



2nd order derivatives are continuous

Clairaut's theorem

Example



Example

Show that the function $u(x,y) = 3x^2y + 2x^2 - y^3 - 2y^2$ is a harmonic function and find its harmonic conjugate $v(x,y)$. Such that $f(z) = u(x,y) + iv(x,y)$ is an analytic function.

$$u(x,y) = 3x^2y + 2x^2 - y^3 - 2y^2$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \qquad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \quad (1)$$

$$\frac{\partial u}{\partial x} = 6xy + 4x$$

$$\Rightarrow \frac{\partial v}{\partial y} = 6xy + 4x$$

$$\Rightarrow V(x,y) = \int (6xy + 4x)\, dy$$

$$= \frac{6x y^2}{2} + 4xy + g(x)$$

$$V(x,y) = 3xy^2 + 4xy + g(x)$$

those variables who vanishes in the differentiation

$$u(x,y) = 3x^2y + 2x^2 - y^3 - 2y^2$$

$$\frac{\partial u}{\partial y} = 3x^2 + 0 - 3y^2 - 2y$$

$$\frac{\partial v}{\partial x} = 6xy + 4y + g'(x)$$

(2) $\Rightarrow$ $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$

$$-3x^2 + 3y^2 + 2y = 6xy + 4y + g'(x)$$

$$\Rightarrow \frac{d}{dx}(f(x)) = -3x^2 + 3y^2 - 2y - 6xy$$

$$\Rightarrow f(x) = \int (-3x^2 + 3y^2 - 2y - 6xy)\, dx$$

$$= -\cancel{3}\frac{x^{\cancel{3}}}{\cancel{3}} + 3y^2 x - 2xy - \cancel{3}y\frac{x^2}{\cancel{2}} + C$$

$$f(x) = -x^3 + 3xy^2 - 2xy - 3x^2y + c$$

$$V(x,y) = 3xy^2 + \cancel{6xy} - x^3 + \cancel{3x^2y} - \cancel{2xy} - 3x^2y + c$$

$$= 6xy^2 + 2xy - x^3 - 3x^2y + c$$

$$f(z) = u(x,y) + i\,v(x,y)$$

Example

Example

Show that the function $u(x,y) = e^{-2x}\sin(2y)$ is a harmonic function and find its harmonic conjugate $v(x,y)$. Such that $f(z) = u(x,y) + iv(x,y)$ is an analytic function.

Example

Example

Show that the function $u(x,y) = \ln(x^2+y^2)$ is a harmonic function and find its harmonic conjugate $v(x,y)$. Such that $f(z) = u(x,y) + iv(x,y)$ is an analytic function.

sin(x)
cos(x)
tan(x)

$0 \le t < 2\pi$

$(\cos(t), \sin(t))$

$x^2 + y^2 = 1$

$x^2 + y^2 = \cos^2 t + \sin^2 t = 1$

sinh(x)
cosh(x)
tanh(x)

$(\cosh(t), \sinh(t))$

$x^2 - y^2 = 1$

$\cosh^2 t - \sinh^2 t = 1$

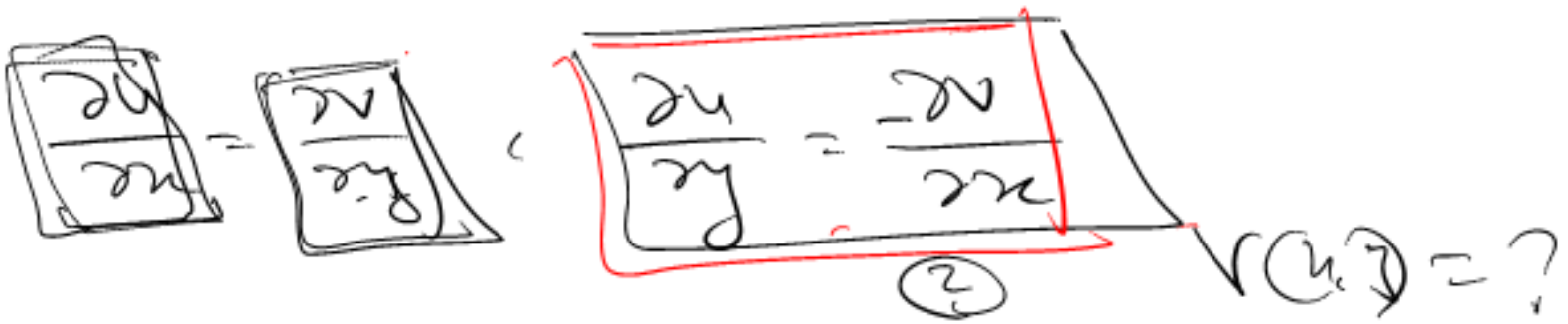
Example

Example

Show that the function $u(x,y) = \cos(x)\sinh(y)$ is a harmonic function and find its harmonic conjugate $v(x,y)$. Such that $f(z) = u(x,y) + iv(x,y)$ is an analytic function.

Hint:

$$\frac{d}{dx}(\sinh x) = \cosh x, \quad \frac{d}{dx}(\cosh x) = \sinh x$$



$$\frac{\partial u}{\partial x} = \frac{\partial}{\partial x}\left(\cos(x)\sinh(y)\right) = \sinh(y)\left(-\sin(x)\right)$$

$$= \boxed{-\sin(x)\sinh(y)}$$

$$\frac{\partial v}{\partial y} = -\sin(x)\sinh(y)$$

$$v(x,y) = -\int \sin(x)\sinh(y)\,dy$$

$$= -\sin(x)\int \sinh(y)\,dy$$

$$= -\sin(x)\cosh(y) + \boxed{g(x)}$$

$$u(x,y) = \cos(x)\sinh(y)$$

$$\frac{\partial u}{\partial y} = \cos(x)\cosh(y)$$

Given,

$$V(x,y) = -\sin(x)\cosh(y) + g(x)$$

$$\frac{\partial V}{\partial x} = -\cos(x)\cosh(y) + g'(x)$$

$$\Rightarrow \quad \frac{\partial u}{\partial y} = -\frac{\partial V}{\partial x}$$

$$\Rightarrow \cos(x)\cosh(y) = +\cos(x)\cosh(y) + g'(x)$$

$$\Rightarrow \quad g'(x) = 0$$

$$\Rightarrow \quad \frac{d}{dx}(g(x)) = 0$$

$$\Rightarrow \quad g(x) = \int 0 \, dx = 0 + C = C$$

$$V(x,y) = -\sin(x)\cosh(y) + C$$

Example

Example

Show that the function $u(x,y) = xe^{-x}\sin(y) - e^{-x}y\cos(y)$ is a harmonic function and find its harmonic conjugate $v(x,y)$. Such that $f(z) = u(x,y) + iv(x,y)$ is an analytic function.